

Data Quality Report

Quality Score: 94%

Missing Values:

index 0 0 Company 0 1 Location_HQ 0 2 Industry 0 3 Laid_Off_Count 1253 4 Date 0 5 Source 0 6
Funds_Raised 390 7 Stage 0 8 Date_Added 0 9 Country 0 10 Percentage 1300 11
List_of_Employees_Laid_Off 0

Duplicates:

{'total_duplicates': np.int64(0)}

Outliers:

{'Laid_Off_Count': 279, 'Funds_Raised': 357, 'Percentage': 304}

AI Insight:

Based on the provided dataset quality metrics, here is a detailed analysis: #### Overall Data Quality Score: 94% - A score of 94% indicates that the dataset is of high quality overall. However, there are areas for improvement, particularly concerning missing values. #### Missing Values - **Laid_Off_Count:** 1,253 missing values - **Funds_Raised:** 390 missing values - **Percentage:** 1,300 missing values - Other columns have no missing values. ##### Analysis: - The presence of missing values, particularly in key metrics like `Laid\_Off\_Count`, `Funds\_Raised`, and `Percentage`, is a concern. This could lead to biases in analysis and affect decision-making. Strategies to handle missing data, such as imputation, removal, or further investigation into the reasons for these missing values, should be considered. #### Duplicates - **Total Duplicates:** 0 ##### Analysis: - The absence of duplicates is positive, as it ensures the integrity of the dataset and that each record is unique. This is crucial for accurate analysis and reporting. #### Outliers - **Laid_Off_Count:** 279 outliers - **Funds_Raised:** 357 outliers - **Percentage:** 304 outliers ##### Analysis: - The presence of outliers in significant metrics such as `Laid\_Off\_Count`, `Funds\_Raised`, and `Percentage` can distort analysis results. It is important to investigate the nature of these outliers: - Are they valid extreme values or errors in data entry? - Do they represent a small number of companies with large layoffs or funding? Handling outliers may require methods such as capping, transformation, or exclusion, depending on their impact on the analysis. #### Recommendations 1. **Address Missing Values:** - Implement strategies to handle missing values, especially for `Laid\_Off\_Count`, `Funds\_Raised`, and `Percentage`. This may include data imputation based on similar records or historical averages. 2. **Investigate Outliers:** - Conduct a deeper analysis into the outliers for `Laid\_Off\_Count`, `Funds\_Raised`, and `Percentage`. Determine if they are valid or need to be adjusted. 3. **Maintain Data Integrity:** - Continue to monitor for duplicates and maintain the process for data entry to avoid future duplicates. 4. **Continuous Quality Assessment:** - Regularly assess data quality and consider implementing automated data validation checks to maintain a high quality score. By addressing these areas, the overall reliability and usability of the dataset can be significantly enhanced, leading to better insights and decision-making.